UChicago REU Notes Week 5

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To be honest, last week's note isn't as successful, so we'll mainly focus on the category theory from now on. Here, we assume that G is a finite group, and the categories (mentioned as objects of Cat or similar objects) are small.

1 Categorical Preliminaries

This part is devoted to understand some basics. I'll fill in the details for preliminaries on G-categories later, together with adjunction ($\mathrm{Ob} \dashv \mathcal{E}$) between Cat and Set.

Given a category $\mathcal{A}$, the functor category $\mathsf{Cat}(\mathcal{E}(\mathcal{G}), \mathcal{A})$ is automatically a G-category, as each element $g \in G$ can be realized as an automorphism functor on the indiscrete category $\mathcal{E}G$, so it admits both a left/right action by precomposing either g^{-1} or g as automorphism of $\mathcal{E}G$.

Similarly, if $\mathcal{A}$ itself already has a left/right G-action, then it defines a left/right G-action on $\mathsf{Cat}(\mathcal{B}, \mathcal{A})$ by postcomposition; similarly if $\mathcal{B}$ is a G-category, then precomposition also defines one. Combining the two, it admits a conjugation action on $\mathsf{Cat}(\mathcal{B}, \mathcal{A})$ if both $\mathcal{B}, \mathcal{A}$ admits a left action.

Given G-categories $\mathcal{A}, \mathcal{B}$, denote $\mathsf{Cat}_G(\mathcal{B}, \mathcal{A}) := \mathsf{Cat}(\mathcal{B}, \mathcal{A})$, and $G\mathsf{Cat}(\mathcal{B}, \mathcal{A}) := \mathsf{Cat}(\mathcal{B}, \mathcal{A})^G$ as the G-fixpoint (i.e. all functors commuting with G-actions).

Proposition 1.1. *There is a G-equivarant isomorphism when giving $\mathcal{A}, \mathcal{B}, \mathcal{C}$ three G-categories:*

$$\Phi : \mathsf{Cat}_G(\mathcal{A} \times \mathcal{B}, \mathcal{C}) \cong \mathsf{Cat}_G(\mathcal{A}, \mathsf{Cat}(\mathcal{B}, \mathcal{C}))$$

Proof. We recognize this as a non-equivariant statement first, then show if all three are G-categories (with G acts diagonally on $\mathcal{A} \times \mathcal{B}$), such isomorphism is G-equivariant.

I. Construction of Functor:

Given a functor $F : \mathcal{A} \times \mathcal{B} \to \mathcal{C}$, for any object $X \in \mathcal{A}$ (together with identity $\mathrm{id}_X : X \to X$), define a functor $F_X : \mathcal{B} \to \mathcal{C}$ by $F_X(Y) := F(X, Y)$, and given any morphism $g : Y \to Y'$ in $\mathcal{B}$, define the morphism $F_X g : F_X(Y) \to F_X(Y')$ as the morphism $F(\mathrm{id}_X, g) : F(X, Y) \to F(X, Y')$. The routine check that this is functorial, using the product category definition:

- Given $\mathrm{id}_Y : Y \to Y$, we have $F_X \mathrm{id}_Y = F(\mathrm{id}_X, \mathrm{id}_Y) = \mathrm{id}_{F(X,Y)} = \mathrm{id}_{F_X(Y)}$.

- Given $g : Y \to Y'$, $g' : Y' \to Y''$, we have

$$F_X(g' \circ g) = F(\mathrm{id}_X, g' \circ g) = F(\mathrm{id}_X, g') \circ F(\mathrm{id}_X, g) = F_X g' \circ F_X g$$

Now, given a morphism $f : X \to X'$ in $\mathcal{A}$, we shall define a natural transformation $Nf : F_X \to F_{X'}$. Realistically, for any $Y \in \mathcal{B}$, the only choice is by $Nf_Y := F(f, \mathrm{id}_Y) : F(X, Y) \to F(X', Y)$. Which, given $g : Y \to Y'$ in $\mathcal{B}$, the following diagram commutes using product category property:

$$\begin{tikzcd} F(X,Y) \arrow[r, "F(\mathrm{id}_X, g)"] \arrow[d, "F(f, \mathrm{id}_Y)"'] & F(X,Y') \arrow[d, "F(f, \mathrm{id}_{Y'})"] & \equiv & F_X(Y) \arrow[r, "F_X g"] \arrow[d, "Nf_Y"'] & F_X(Y') \arrow[d, "Nf_{Y'}"] \\ F(X',Y) \arrow[r, "F(\mathrm{id}_{X'}, g)"'] & F(X',Y') & & F_{X'}(Y) \arrow[r, "F_{X'} g"'] & F_{X'}(Y') \end{tikzcd}$$

Hence, $Nf : F_X \to F_{X'}$ is a natural transformation.

Then, we claim that $N : \mathcal{A} \to \mathsf{Cat}(\mathcal{B}, \mathcal{C})$ by $N(X) := F_X$, $Nf : F_X \to F_{X'}$ as above is a functor:

- $(N\mathrm{id}_X)_Y = F(\mathrm{id}_X, \mathrm{id}_Y) = \mathrm{id}_{F(X,Y)} = \mathrm{id}_{F_X(Y)}$, so $N\mathrm{id}_X = \mathrm{id}_{F_X}$.
- Given $f : X \to X'$, $f' : X' \to X''$, then

$$N(f' \circ f)_Y = F(f' \circ f, \mathrm{id}_Y) = F(f', \mathrm{id}_Y) \circ F(f, \mathrm{id}_Y) = Nf'_Y \circ Nf_Y$$

  So, $N(f' \circ f) = Nf' \circ Nf$.

Now, suppose two such functors $F, F' : \mathcal{A} \times \mathcal{B} \to \mathcal{C}$ has a natural transformation $\mu : F \to F'$, we want to construct a natural transformation $\overline{\mu} : N \to N'$, where $N, N' : \mathcal{A} \to \mathsf{Cat}(\mathcal{B}, \mathcal{C})$ are functors associated to F, F'. Which, given any $X \in \mathcal{A}$, we see the collection $(\overline{\mu}_X)_Y := \mu_{(X,Y)}$ is a natural transformation $F_X \to F'_X$, due to the following diagram for all $g : Y \to Y'$ in $\mathcal{B}$:

$$\begin{tikzcd} F(X,Y) \arrow[r, "F(\mathrm{id}_X, g)"] \arrow[d, "\mu_{(X,Y)}"'] & F(X,Y') \arrow[d, "\mu_{(X,Y')}"] & \equiv & F_X(Y) \arrow[r, "F_X g"] \arrow[d, "(\overline{\mu}_X)_Y"'] & F(X,Y') \arrow[d, "(\overline{\mu}_X)_{Y'}"] \\ F'(X,Y) \arrow[r, "F'(\mathrm{id}_X, g)"'] & F'(X,Y') & & F'_X(Y) \arrow[r, "F'_X g"'] & F'_X(Y') \end{tikzcd}$$

This shows $\overline{\mu}_X : F_X \to F'_X$ is a natural transformation, or morphism within $\mathsf{Cat}(\mathcal{B}, \mathcal{C})$. Then, notice that if $\mu : F \to F'$, $\mu' : F' \to F''$ are two natural transformations between functors $F, F', F'' : \mathcal{A} \times \mathcal{B} \to \mathcal{C}$, the following holds:

$$(\overline{\mu' \circ \mu}_X)_Y = (\mu' \circ \mu)_{(X,Y)} = \mu'_{(X,Y)} \circ \mu_{(X,Y)} = (\overline{\mu'}_X)_Y \circ (\overline{\mu}_X)_Y$$

This shows as natural transformations, $\overline{\mu}_X : N_X \to N'_X$ and $\overline{\mu'}_X : N'_X \to N''_X$ satisfies $(\overline{\mu' \circ \mu})_X = \overline{\mu'}_X \circ \overline{\mu}_X$, so $\overline{\mu' \circ \mu} = \overline{\mu'} \circ \overline{\mu}$ as natural transformations $\overline{\mu} : N \to N'$, $\overline{\mu'} : N' \to N''$.

So, define $\Phi(F : \mathcal{A} \times \mathcal{B} \to \mathcal{C}) := N : \mathcal{A} \to \mathsf{Cat}(\mathcal{B}, \mathcal{C})$, and $\Phi(\mu : F \to F') := \overline{\mu} : N \to N'$ as above. $\square$